High-Resolution Forward-Looking Multichannel SAR Imagery With Array Deviation Angle Calibration

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***Abstract*— Traditional synthetic aperture radar (SAR) imaging is limited to achieve the high-resolution image of the side-looking areas. Nevertheless, equipped with a small size linear array across the trajectory, forward-looking multichannel SAR (FLMC-SAR) is capable of reconstructing the high-resolution image of the front area. In FLMC-SAR imaging framework, the left-right Doppler ambiguity is expected to resolve with beamforming approaches using the multichannel system diversity. However, beamforming-based Doppler ambiguity resolving is sensitive to the array deviation angle, which causes a mismatch between the azimuth angle and Doppler frequency. In this article, we propose an array deviation angle calibration and imagery algorithm for FLMC-SAR. The space–time characteristic of FLMC-SAR is explored and the range-dependent array deviation angle model is established. Following the Doppler beam sharpening imaging, strong targets are selected to derive the mismatch of the space–time characteristic. A maximum likelihood estimation of the array deviation angle is developed to modify the matching between the azimuth angle and Doppler frequency. Therefore, the left-right Doppler ambiguity can be solved correctly, yielding high-resolution FLMC-SAR imagery. Extensive simulation and real data experiments are performed to demonstrate the effectiveness of the proposed method.**

***Index Terms*— Array deviation angle calibration, beamforming, forward-looking multichannel synthetic aperture radar (FLMC-SAR), left-right Doppler ambiguity, space–time characteristic.**

I. Introduction

SYNTHETIC aperture radar (SAR) [1]–[4] can generate high-resolution microwave images, regardless of weather conditions, leading to extensive applications in practical engineering, such as aircrafts, civilian remote sensing, and so on. Traditional SAR usually has its observation area on the side or squint side of the trajectory. As such traditional SAR algorithms are frequently used in various modes such as side-looking SAR and squint-looking SAR. Limited by this imaging principle, however, all of these models refuse to focus the targets ahead of the flight direction, resulting in a forward-looking blind zone. Forward-looking SAR, on the other hand, places the imaging area directly in front of the trajectory. Its characteristic geometry mode produces a limited Doppler diversity, placing restrictions on the azimuth resolution. Furthermore, under the specific geometric model of forward-looking imaging, the targets that are symmetrical about the trajectory have the same slant range history, resulting in the left-right Doppler ambiguity across the trajectory.

Forward-looking SAR is an innovative system. The existing forward-looking imaging systems can be grouped into the following categories.

1) German Aerospace Center first put forward the Sector Imaging Radar for Enhanced Vision (SIREV) [5], [6] system in the forward-looking mode and also did some primary simulation researches. Based on this, some related research works, such as chirp scaling algorithm [7], [8], compressive sensing algorithm [9], and 3-D imaging algorithm [10], [11], are suitable for the conventional SIREV system. However, the system's azimuth resolution is determined by the real aperture of the array. In order to obtain a higher forward-looking resolution, the system usually needs a large array antenna, which leads to increasing the complexity of the system.
2) The Bistatic Forward-Looking Synthetic Aperture Radar [12], [13] (BFSAR) is also an important forward-looking imaging system. For a BFSAR system, the platform motion error is a problem which arises from the trajectory deviation and instability of platform velocity. Some motion compensation and autofocus algorithms [14]–[16] are proposed to improve imaging quality of BFSAR, leading to some applications in aviation [17]–[19]. The BFSAR system is usually equipped with a forward-looking receiver and a side looking transmitter, both of which move along the same direction at the same velocity. That is, the imaging area is often located in front of the receiver and on the side of the transmitter. The receiver achieves forward-looking imaging through the synthetic aperture gain provided by the side looking transmitter. For the transmitter, there is a limitation in forward-looking imaging.
3) Multichannel SAR as an important tool of forward-looking imaging has become a hot research topic [20]–[23]. With the dense array, multichannel SAR is capable

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of imaging for the forward-looking area [24], [25]. In [26], a novel imaging method of forward-looking multichannel SAR (FLMC-SAR) was proposed to reconstruct the image of the forward-looking area and then utilize beamforming to solve the left-right Doppler ambiguity. Compared with other systems having the dense array, the FLMC-SAR system utilizes synthetic apertures for imaging and its multichannel system for eliminating the left-right Doppler ambiguity so that a small size array can be utilized to accomplish forward-looking imaging.

The left-right Doppler ambiguity is an inherent problem that has to be faced in FLMC-SAR imaging [27]. Beamforming is an important tool to solve the left-right Doppler ambiguity in FLMC-SAR imaging via maintaining the signal in the desired target direction while suppressing that in the ambiguous direction. However, beamforming becomes very sensitive to space–time characteristic, which is determined by the azimuth angle and Doppler frequency. Hence, the mismatch between the azimuth angle and Doppler frequency can easily lead to the failure of eliminating the left-right Doppler ambiguity. The mismatch problem is usually caused by trajectory deviation and array errors [28]–[32]. In practical applications, the deviation of radar from the ideal trajectory results in motion errors and thus produces phase errors in the signal history, finally leading to defocusing. And, the Doppler shift induced by these phase errors degrades the performance of Doppler ambiguity suppression in the spatial domain. In order to eliminate such phase errors, the deviation should be measured through a navigation system and then compensated [33], [34]. On the other hand, the array errors would become an obstacle on the way to resolve the left-right Doppler ambiguity. However, the problem of performance degradation in the Doppler ambiguity resolving caused by spatial domain array errors is difficult to solve through the motion compensation alone.

The conventional array calibration method, such as the rotating element electric field vector (REV) [35], [36] algorithm, is effective to correct the array gain and phase error. In [37], a maximum-likelihood calibration algorithm is presented to compensate for the array gain, the phase error, as well as the array position errors. Although these techniques show good performance in array calibration, they require a set of calibration sources in known locations, which is not applicable in FLMC-SAR imaging.

Both motion compensation and array calibration are significant in FLMC-SAR imaging, but there are some differences between them. For FLMC-SAR imaging, we first carry out SAR imaging in the time domain and then solve the left-right Doppler ambiguity in the spatial domain. Motion compensation is mainly used to compensate the phase error of signals to improve the resolution of SAR imaging in the time domain, while the array calibration aims at correcting the steering vector of targets to improve the performance of the beamforming-based left-right Doppler ambiguity resolving in the spatial domain.

Motivated by the aforementioned problems in FLMC-SAR, we propose a space–time adaptive array deviation angle calibration method to solve the mismatch problem between azimuth angle and Doppler frequency for removing the FLMC-SAR Doppler ambiguity. First, the geometry model of FLMC-SAR is discussed. Second, the left-right Doppler ambiguity in the FLMC-SAR system is analyzed. In the ideal case, there is no mismatch between azimuth angle and Doppler frequency of the ambiguous targets, and an unambiguous FLMC-SAR image can be obtained by beamforming. In the actual work, however, the array deviation angle will cause the real space–time spectrum to deviate from the ideal one, leading to a degradation in the performance of removing the left-right Doppler ambiguity. To cope with this problem, we investigate the array deviation angle, which can be described by an angle between the real array structure and the ideal one. Moreover, we discuss this space–time mismatch influence on the left-right Doppler ambiguity removal. In order to embed the adaptive calibration of the array deviation angle into the FLMC-SAR imaging process, the space–time cascading processing is introduced. At first, time-domain processing is performed with the Doppler Beam Sharpening (DBS) to obtain the ambiguous imaging result. After that, space–time processing is followed, strong and dominated targets are selected to derive the mismatch of the space–time characteristics, and maximum likelihood estimation of the array deviation angle is developed. Correcting the mismatch between the azimuth angle and Doppler frequency in the spatial domain, we can optimally resolve the left-right Doppler ambiguity via beamforming to achieve FLMC-SAR imagery. The proposed algorithm has some innovations as follows.

1) For the first time, the array deviation angle is regarded as a key factor in modeling the array structure and is utilized to modify the matching between azimuth angle and Doppler frequency in order to remove the left-right Doppler ambiguity.
2) The array deviation angle calibration is realized by space–time cascading processing. The mismatch between the azimuth angle and Doppler frequency is corrected by a maximum likelihood estimation of the array deviation angle. The DBS algorithm is performed in the time domain to obtain the ambiguous imaging result. This avoids the multitarget angle estimation in the same beam, thus improving the array deviation angle estimation. In addition, the proposed imaging method carries out SAR imaging in the time domain and then solves the left-right Doppler ambiguity in the spatial domain. The space–time cascaded array deviation angle calibration method is extremely compatible with imaging processing.
3) To cope with the range-dependent array deviation angle, we propose a data-driven adaptive array deviation angle calibration method. Data-driven adaptive processing is not only able to ensure the accuracy of the array deviation angle estimation but also can be integrated with the mismatch correction of space–time characteristics. This mismatch correction is the key to improve the beamforming-based left-right Doppler ambiguity resolving.

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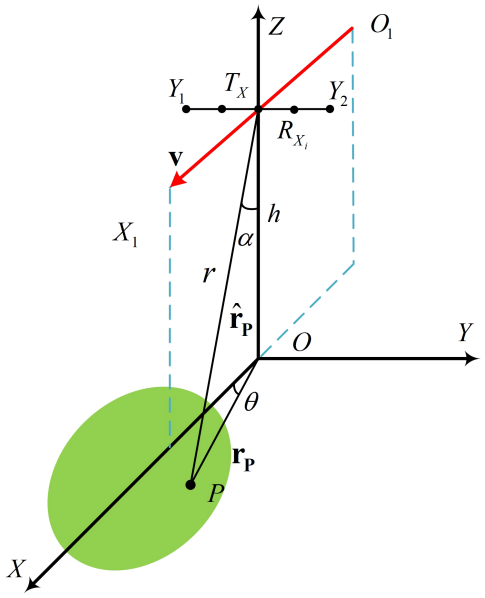


Fig. 1. Geometry model of spotlight FLMC-SAR.

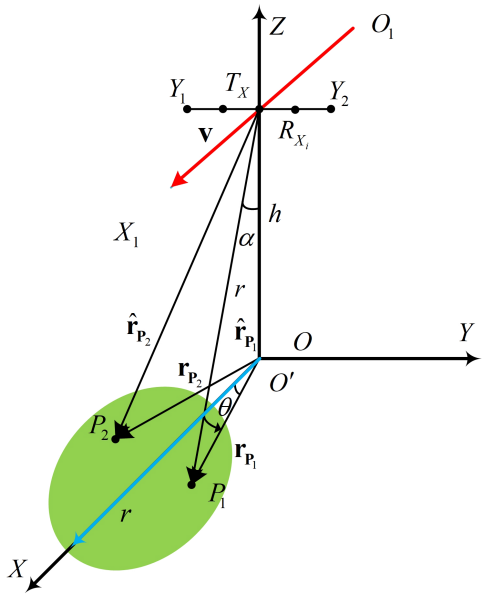


Fig. 2. Geometry model with two symmetric targets.

This article is organized as follows. Section II gives the signal and geometry model of FLMC-SAR and details the left-right Doppler ambiguity problem. A beamforming-based left-right Doppler ambiguity resolving algorithm for FLMC-SAR is introduced. Section III presents the array structure and space–time adaptive array deviation angle calibration. In Section IV, the experimental and real imaging results are analyzed. In Section V, some crucial conclusions of this article are drawn.

II. FLMC-SAR System Model

A. FLMC-SAR Geometry Model

The geometry model of spotlight FLMC-SAR is shown in Fig. 1. In the Cartesian coordinate system $OXYZ$, we set the projection of the aperture center on the ground as the origin O and the direction of the nominal trajectory as X-axis. The straight line O_1X_1 along the X-axis denotes the nominal trajectory with a height of h. The system adopts the linear array antenna as the receiving antenna, which is symmetrically distributed along the Y-axis over Y_1Y_2 provided by R_{X_i}. The length of the antenna's real aperture is D, and the antenna element spacing is d. T_X is the transmitting element, which is located at the midpoint of Y_1Y_2 and coincident with the central receiving antenna.

In an ideal case, the antenna phase center (APC) of the transmitter and receiver is assumed to move along the nominal trajectory O_1X_1 and its velocity is $\mathbf{v}[v, 0, 0]$. At an instantaneous radar slow-time t_m, the simultaneous positions of the transmitter and receiver are defined as $\mathbf{T_X}[vt_m, 0, h]$ and $\mathbf{R_{X_i}}[vt_m, y_i, h]$, respectively, where y_i denotes the azimuth coordinate of the ith receiving antenna.

The symbol P denotes an arbitrary target located at the imaging area. We define the vector pointing toward the target from the aperture center as the radar's light of sight $\hat{\mathbf{r}}_P[r\sin\alpha\cos\theta, r\sin\alpha\sin\theta, -h]$, which is variant with the target coordinate. The angle between the radar's light of sight and the Z-axis is the beam pointing pitch angle α. The angle between the projection of the radar's light of sight on the ground and the X-axis is the beam pointing azimuth angle θ. The range between the target and the APC at the aperture center is the reference slant range r of the target. Assuming $\mathbf{r_P}[r\sin\alpha\cos\theta, r\sin\alpha\sin\theta, 0]$ is the position of the point target with the beam pointing pitch angle α, the beam pointing azimuth angle θ, and the reference slant range r, the ideal instantaneous range $R_T(R_R)$ from the target to the transmitting (receiving) is given by

$$\begin{aligned} R_T(t_m; r, \theta, 0) &= |\mathbf{T_X} - \mathbf{r_P}| \\ &= \sqrt{(vt_m - r\sin\alpha\cos\theta)^2 + (r\sin\alpha\sin\theta)^2 + h^2} \end{aligned} \tag{1}$$

$$\begin{aligned} R_R(t_m; r, \theta, y_i) &= |\mathbf{R_{X_i}} - \mathbf{r_P}| \\ &= \sqrt{(vt_m - r\sin\alpha\cos\theta)^2 + (y_i - r\sin\alpha\sin\theta)^2 + h^2}. \end{aligned} \tag{2}$$

B. Left-Right Doppler Ambiguity in FLMC-SAR

As shown in Fig. 2, for the simplicity of left-right Doppler ambiguity analysis, we define the targets' coordinate in a polar coordinate system (PCS) in the scene area. Also, the point O' is the origin of the PCS, which is located at the origin of the Cartesian coordinate system $OXYZ$, and the polar angle rotates counterclockwise. The two arbitrary targets P_1 and P_2 are symmetric about the X-axis. Their coordinates in PCS can be expressed as $P_1(\rho, -\theta)$ and $P_2(\rho, \theta)$, where ρ denotes the projection of radar's light of sight on the ground and θ denotes the beam pointing azimuth angle. One can notice that P_1 and P_2 have the same beam pointing pitch angle α, the same reference slant range r, but the opposite beam pointing azimuth angle $-\theta$ and θ. And their coordinates in $OXYZ$ are $P_1(r\sin\alpha\cos\theta, -r\sin\alpha\cos\theta, 0)$ and $P_2(r\sin\alpha\cos\theta, r\sin\alpha\cos\theta, 0)$, respectively. The two-way

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slant range of the symmetric targets is

$$\begin{aligned} &R_{P_1}(t_m; r, -\theta, y_i) \\ &\quad = R(t_m; r, -\theta, y_i) \\ &\quad = R_T(t_m; r, -\theta, 0) + R_R(t_m; r, -\theta, y_i) \\ &\quad = \sqrt{(vt_m - r\sin\alpha\cos\theta)^2 + (r\sin\alpha\sin\theta)^2 + h^2} \\ &\qquad + \sqrt{(vt_m - r\sin\alpha\cos\theta)^2 + (y_i + r\sin\alpha\sin\theta)^2 + h^2} \end{aligned} \tag{3}$$

$$\begin{aligned} &R_{P_2}(t_m; r, \theta, y_i) \\ &\quad = R(t_m; r, \theta, y_i) \\ &\quad = R_T(t_m; r, \theta, 0) + R_R(t_m; r, \theta, y_i) \\ &\quad = \sqrt{(vt_m - r\sin\alpha\cos\theta)^2 + (r\sin\alpha\sin\theta)^2 + h^2} \\ &\qquad + \sqrt{(vt_m - r\sin\alpha\cos\theta)^2 + (y_i - r\sin\alpha\sin\theta)^2 + h^2}. \end{aligned} \tag{4}$$

It follows that the point targets P_1 and P_2 with X-axis symmetry have the same transmitting slant range but different receiving slant range. The receiving slant range between P_1 and P_2 is changed with the variation of the coordinate of the receiving antenna y_i. The difference between (3) and (4) is

$$\begin{aligned} &R_{P_1}(t_m; r, -\theta, y_i) - R_{P_2}(t_m; r, \theta, y_i) \\ &\quad = R_T(t_m; r, -\theta, 0) + R_R(t_m; r, -\theta, y_i) \\ &\qquad -(R_T(t_m; r, \theta, 0) + R_R(t_m; r, \theta, y_i)) \\ &\quad = R_R(t_m; r, -\theta, y_i) - R_R(t_m; r, \theta, y_i) \\ &\quad = \sqrt{(vt_m - r\sin\alpha\cos\theta)^2 + (y_i + r\sin\alpha\sin\theta)^2 + h^2} \\ &\qquad - \sqrt{(vt_m - r\sin\alpha\cos\theta)^2 + (y_i - r\sin\alpha\sin\theta)^2 + h^2} \\ &\quad = \sqrt{{R_0}^2(t_m) + (y_i + r\sin\alpha\sin\theta)^2} \\ &\qquad - \sqrt{{R_0}^2(t_m) + (y_i - r\sin\alpha\sin\theta)^2} \end{aligned} \tag{5}$$

where $R_0(t_m) = ((vt_m - r\sin\alpha\cos\theta)^2 + h^2)^{1/2}$. Let ΔR be the range resolution. With the relation of $y_i \ll R_0(t_m)$, the following approximations are valid:

$$\begin{aligned} &R_{P_1}(t_m; r, -\theta, y_i) - R_{P_2}(t_m; r, \theta, y_i) \\ &\qquad \approx \frac{2r\sin\alpha\sin\theta}{\sqrt{R_0(t_m) + (r\sin\alpha\sin\theta)^2}} \cdot y_i \ll \Delta R. \end{aligned} \tag{6}$$

For the two point targets that are symmetrical about the trajectory, there is a difference between the two-way slant ranges of each channel, but the difference is too small to cause range cell migration. Therefore, the echo signal of the symmetric target will be accumulated in the same range bin, causing the left-right Doppler ambiguity of FLMC-SAR. Although the symmetric targets are ambiguous on the SAR imaging plane, they have different azimuthal angles, which results in different steering vectors relative to the array antenna. In FLMC-SAR imaging, the Doppler ambiguity exists only when targets are left-right symmetrical. We first perform the DBS algorithm in the time domain to obtain the ambiguous imaging result. Since synthetic aperture provides high azimuth resolution, ambiguous targets with different azimuth angles have been distinguished after DBS imaging. And we can obtain the ambiguous result of the left-right symmetrical targets. Then the beamforming is performed to achieve Doppler ambiguity resolving. Although the targets are within one beamwidth, there are still differences in the steering vector of the left-right symmetrical targets. So we can get the unambiguous imaging result via beamforming.

C. Left-Right Doppler Ambiguity Resolving via Beamforming

Assuming that the linear frequency modulated (LFM) signal with wavelength λ is transmitted, the signal of the two point targets that are symmetrical about the trajectory can be obtained as follows:

$$\begin{aligned} &s_{P_1}(\hat{t}, t_m; r, -\theta, y_i) \\ &\quad = A_{P_1} w_r\left(\hat{t} - \frac{R(t_m; r, -\theta, y_i)}{c}\right) \\ &\qquad \exp\left[\mathrm{j}\pi\gamma\left(\hat{t} - \frac{R(t_m; r, -\theta, y_i)}{\mathrm{c}}\right)^2\right] \\ &\qquad \exp\left[-j\frac{2\pi}{\lambda}R(t_m; r, -\theta, y_i)\right] \text{ s.t. } -\frac{D}{2} \le y_i \le \frac{D}{2} \end{aligned} \tag{7}$$

$$\begin{aligned} &s_{P_2}(\hat{t}, t_m; r, \theta, y_i) \\ &\quad = A_{P_2} w_r\left(\hat{t} - \frac{R(t_m; r, \theta, y_i)}{c}\right) \\ &\qquad \exp\left[\mathrm{j}\pi\gamma\left(\hat{t} - \frac{R(t_m; r, \theta, y_i)}{\mathrm{c}}\right)^2\right] \\ &\qquad \exp\left[-j\frac{2\pi}{\lambda}R(t_m; r, \theta, y_i)\right] \text{ s.t. } -\frac{D}{2} \le y_i \le \frac{D}{2}. \end{aligned} \tag{8}$$

Due to the left-right Doppler ambiguity, the signal of the targets will be accumulated in the same range and Doppler bin. And it can be described by (9).

$$\begin{aligned} &s_0(\hat{t}, t_m; r, \theta, y_i) \\ &= s_0(\hat{t}, t_m; r, -\theta, y_i) \\ &= s_{P_1}(\hat{t}, t_m; r, -\theta, y_i) + s_{P_2}(\hat{t}, t_m; r, \theta, y_i)] \\ &= A_{P_1} w_r\left(\hat{t} - \frac{R(t_m; r, -\theta, y_i)}{c}\right) \\ &\qquad \cdot \exp\left[\mathrm{j}\pi\gamma\left(\hat{t} - \frac{R(t_m; r, -\theta, y_i)}{\mathrm{c}}\right)^2\right] \\ &\qquad \cdot \exp\left[-j\frac{2\pi}{\lambda}R(t_m; r, -\theta, y_i)\right] + A_{P_2} w_r\left(\hat{t} - \frac{R(t_m; r, \theta, y_i)}{c}\right) \\ &\qquad \cdot \exp\left[\mathrm{j}\pi\gamma\left(\hat{t} - \frac{R(t_m; r, \theta, y_i)}{\mathrm{c}}\right)^2\right] \\ &\qquad \cdot \exp\left[-j\frac{2\pi}{\lambda}R(t_m; r, \theta, y_i)\right] \\ &\approx w_r\left(\hat{t} - \frac{R(t_m; r, \theta, y_i)}{c}\right) \cdot \exp\left[\mathrm{j}\pi\gamma\left(\hat{t} - \frac{R(t_m; r, \theta, y_i)}{\mathrm{c}}\right)^2\right] \\ &\qquad \cdot \left\{A_{P_1} \cdot \exp\left[-j\frac{2\pi}{\lambda}R(t_m; r, -\theta, y_i)\right] + A_{P_2} \right. \\ &\qquad \left. \cdot \exp\left[-j\frac{2\pi}{\lambda}R(t_m; r, \theta, y_i)\right]\right\} \text{ s.t. } -\frac{D}{2} \le y_i \le \frac{D}{2}. \end{aligned} \tag{9}$$

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In order to achieve unambiguous imaging, we propose an algorithm via beamforming. First, the signal is processed by the DBS algorithm to achieve the imaging with ambiguity, then the beamforming is performed to solve the left-right Doppler ambiguity.

The first step of the conventional image algorithm, such as the range Doppler algorithm, and chirp scaling algorithm, is to remove the RCM caused by the ideal data acquisition geometry. Since the ideal RCM is deterministic, it can be accurately corrected. The signal is subjected to range pulse compression processing, and the pulse compressed signal can be expressed as

$$\begin{aligned} &s_{\mathrm{rc}}(\hat{t}, t_m; r, \theta, y_i) \\ &= s_{\mathrm{rc}}(\hat{t}, t_m; r, -\theta, y_i) \\ &= \mathrm{IFFT}_{\hat{t}}\left\{\mathrm{FFT}_{\hat{t}}\left[s_0(\hat{t}, t_m; r, \theta, y_i)\right]\mathrm{FFT}_{\hat{t}}\left[s_{\mathrm{ref}}(\hat{t})\right]\right\} \\ &= \mathrm{sinc}\left[B\left(\hat{t}-\frac{2r}{c}\right)\right] \\ &\quad \cdot\left\{A_{P_1}\cdot\exp\left[-j\frac{2\pi}{\lambda}R(t_m; r, -\theta, y_i)\right]\right. \\ &\quad \left.+A_{P_2}\cdot\exp\left[-j\frac{2\pi}{\lambda}R(t_m; r, \theta, y_i)\right]\right\} \ \text{s.t.}-\frac{D}{2}\le y_i\le\frac{D}{2} \end{aligned} \tag{10}$$

where the symbol B denotes the signal bandwidth, sinc denotes the sampling function, and $s_{\mathrm{ref}}(\hat{t})$ is the matching filter function, which can be expressed as

$$s_{\mathrm{ref}}(\hat{t}) = w_r(\hat{t})\exp(-j\pi\gamma\hat{t}^2). \tag{11}$$

The Taylor series of the slant range equation between the target and the central receiving antenna is

$$R(t_m; r, -\theta, 0) = 2r + 2v\sin\alpha\cos\theta\cdot t_m + \cdots. \tag{12}$$

Then, (3) is approximately equal to

$$\begin{aligned} &R(t_m; r, -\theta, y_i) \\ &\quad = R(t_m; r, -\theta, 0) + (R(t_m; r, -\theta, y_i) - R(t_m; r, -\theta, 0)) \\ &\quad \approx 2r + 2v\sin\alpha\cos\theta\cdot t_m \\ &\qquad + (R(t_m; r, -\theta, y_i) - R(t_m; r, -\theta, 0)). \end{aligned} \tag{13}$$

And (10) can be rewritten as (14).

$$\begin{aligned} &s_{\mathrm{rc}}(\hat{t}, t_m; r, \theta, y_i) \\ &= \mathrm{sinc}\left[\mathrm{B}\left(\hat{\mathrm{t}}-\frac{2\mathrm{r}}{\mathrm{c}}\right)\right] \\ &\quad \cdot\left\{A_{P_1}\cdot\exp\left[-j\frac{2\pi}{\lambda}(2r+2v\sin\alpha\cos\theta\cdot t_m)\right]\right. \\ &\qquad \cdot\exp\left[-j\frac{2\pi}{\lambda}(R(t_m; r, -\theta, y_i) - R(t_m; r, -\theta, 0))\right] \\ &\qquad +A_{P_2}\cdot\exp\left[-j\frac{2\pi}{\lambda}(2r+2v\sin\alpha\cos\theta\cdot t_m)\right] \\ &\qquad \left.\cdot\exp\left[-j\frac{2\pi}{\lambda}(R(t_m; r, \theta, y_i) - R(t_m; r, \theta, 0))\right]\right\} \\ &= \mathrm{sinc}\left[B\left(\hat{t}-\frac{2r}{c}\right)\right]\cdot\exp\left[-j\frac{2\pi}{\lambda}(2v\sin\alpha\cos\theta\cdot t_m)\right] \\ &\quad \cdot\exp\left[-j\frac{2\pi}{\lambda}2r\right] \end{aligned}$$

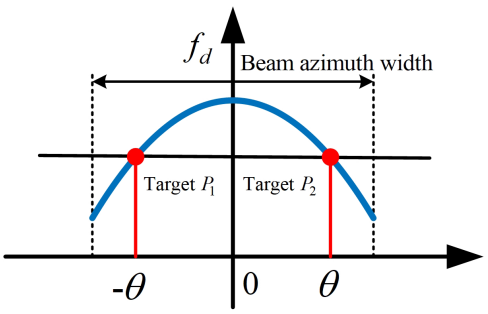


Fig. 3. Space–time spectrum of ambiguous targets.

$$\begin{aligned} &\cdot\left\{A_{P_1}\cdot\exp\left[-j\frac{2\pi}{\lambda}(R(t_m; r, -\theta, y_i) - R(t_m; r, -\theta, 0))\right]\right. \\ &\quad \left.+A_{P_2}\cdot\exp\left[-j\frac{2\pi}{\lambda}(R(t_m; r, \theta, y_i) - R(t_m; r, \theta, 0))\right]\right\} \\ &\approx \mathrm{sinc}\left[B\left(\hat{t}-\frac{2r}{c}\right)\right]\cdot\exp\left[-j\frac{2\pi}{\lambda}(2v\sin\alpha\cos\theta\cdot t_m)\right] \\ &\quad \cdot\exp\left[-j\frac{2\pi}{\lambda}2r\right]\cdot\left\{A_{P_1}\cdot\exp\left[-j\frac{2\pi}{\lambda}(y_i\sin(-\theta))\right]\right. \\ &\qquad \left.+A_{P_2}\cdot\exp\left[-j\frac{2\pi}{\lambda}(y_i\sin\theta)\right]\right\}. \end{aligned} \tag{14}$$

Equation (14) can be divided into four terms. The first term of (14) $\mathrm{sinc}\left[B\left(\hat{t}-(2r/c)\right)\right]$ is the envelope of the signal. The second term of (14) $\exp[-j(2\pi/\lambda)(2v\sin\alpha\cos\theta\cdot t_m)]$ is Doppler frequency, which indicates the real azimuth position of the target, can be expressed as

$$f_d(\theta) = \frac{2v\sin\alpha\cos\theta}{\lambda}. \tag{15}$$

From (15), the space–time correspondence between azimuth angle and Doppler frequency is obtained, as shown in Fig. 3. One can notice that the Doppler frequency is the function of the target's azimuth angle θ. And the symmetric targets P_1 and P_2 owe the same Doppler frequency but the opposite azimuth angles, i.e., the angles with the same absolute value but with opposite signs. It results in the Doppler ambiguity in the FLMC-SAR imaging.

The third term of (14) $\exp[-j(2\pi/\lambda)2r]$ is a constant related to the reference slant range of the target. And the fourth term of (14) $\{A_{P_1}\cdot\exp[-j(2\pi/\lambda)(y_i\sin(-\theta))] + A_{P_2}\cdot\exp[-j(2\pi/\lambda)(y_i\sin\theta)]\}$ is the sum of the complex scattering coefficients accumulated by the symmetric targets. And the symmetric targets present difference in their spatial steering vector, which is of great importance to resolve the followed-up left-right Doppler ambiguity via beamforming.

Applying the azimuth Fourier transform to (14) yields the following imaging result:

$$\begin{aligned} &\mathrm{S}_{rc}(\hat{t}, f_d; r, \theta, y_i) \\ &= \mathrm{FFT}_{t_m}\left[s_{\mathrm{rc}}(\hat{t}, t_m; r, \theta, y_i)\right] \\ &= \mathrm{sinc}\left[B\left(\hat{t}-\frac{2r}{c}\right)\right]\cdot\zeta(f_d - \frac{2v\sin\alpha\cos\theta}{\lambda}) \end{aligned}$$

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$$\cdot \exp\left[-j\frac{2\pi}{\lambda}2r\right]\cdot\left\{A_{P_1}\cdot\exp\left[-j\frac{2\pi}{\lambda}(y_i\sin(-\theta))\right] + A_{P_2}\cdot\exp\left[-j\frac{2\pi}{\lambda}(y_i\sin\theta)\right]\right\} \tag{16}$$

where $\mathrm{FFT}_{t_m}[\cdot]$ is the azimuth Fourier transform operator and $\zeta(f)$ is a shock response function. The imaging result of the target with a reference slant range r and an azimuth angle θ is the peak value of (16), which can be expressed as

$$\begin{aligned}\mathrm{S}_{rc}&\left(\frac{2r}{c}, f_d(\theta); r, \theta, y_i\right)\\ &= \begin{bmatrix}A_{P_1} & A_{P_2}\end{bmatrix}\cdot\begin{bmatrix}\exp\left[-j\frac{2\pi}{\lambda}(y_i\sin(-\theta))\right]\\ \exp\left[-j\frac{2\pi}{\lambda}(y_i\sin\theta)\right]\end{bmatrix}\cdot\exp\left[-j\frac{2\pi}{\lambda}2r\right]\\ &= \begin{bmatrix}A_{P_1} & A_{P_2}\end{bmatrix}\cdot\begin{bmatrix}V(-\theta)\\ V(\theta)\end{bmatrix}\cdot\exp\left[-j\frac{2\pi}{\lambda}2r\right]\\ &= \mathbf{A}\cdot\mathbf{V}\cdot A_{P_0}.\end{aligned} \tag{17}$$

The symbol $\mathbf{A}$ denotes the scattering coefficient matrix of the targets and $\mathbf{V}$ denotes the steering vector matrix. A_{P_0} is a constant, related to the reference slant range of the target. Taking the array center element as the reference element, the array steering vector is given by

$$V(\theta) = \left[\exp\left(j\frac{2\pi}{\lambda}y_1\sin(\theta)\right)\exp\left(j\frac{2\pi}{\lambda}y_2\sin(\theta)\right)\cdots\exp\left(j\frac{2\pi}{\lambda}y_K\sin(\theta)\right)\right] \tag{18}$$

where K is the number of array elements, θ is the beam pointing azimuth angle of the target. With the ambiguous imaging result, an advanced algorithm for left-right Doppler ambiguity via adaptive beamforming is proposed. In the process of beamforming, we use the linearly constrained minimum variance (LCMV) algorithm with two constraints. First, the component of the filtered signal in the desired target direction should be maintained. Second, the component of the filtered signal in the ambiguous direction should be suppressed. The LCMV algorithm can be formulated as follows [38]:

$$\begin{cases}\min_{W_n}\mathbf{W}_n^H\mathbf{R}_X\mathbf{W}_n\\ \text{s.t. } \mathbf{W}_n^H\mathbf{V} = \mathbf{g}_n^H, \quad n = 1,\ldots,N\end{cases} \tag{19}$$

where $\mathbf{W}_n$ is the weight to be determined, the natural number N is the dimension of the left-right Doppler ambiguity in FLMC-SAR imaging, and $\mathbf{g}_n$ is the nth column of $N \times N$ identity matrix. Given the signal received by the array X, the sample covariance matrix $\mathbf{R}_X$, and the steering vector matrix $\mathbf{V}$, the optimal weight of the beamforming filter is determined via the method of Lagrange multipliers [39]

$$\mathbf{W}_n = \mathbf{R_X}^{-1}\mathbf{V}(\mathbf{V}^H\mathbf{R_X}^{-1}\mathbf{V})^{-1}\mathbf{g}_n. \tag{20}$$

With the optimal weight of the beamforming filter, beamforming is accomplished by weighting ambiguity imaging result of each channel and then accumulating them coherently to achieve imaging without Doppler ambiguity. For the left-right Doppler ambiguity problem of FLMC-SAR imaging, assume the dimension of the left-right Doppler ambiguity as $N = 2$. Let $\mathbf{W} = \begin{bmatrix}\mathbf{W}_1 & \mathbf{W}_2\end{bmatrix}$, then the scattering coefficient matrix can be obtained as follows:

$$\begin{aligned}\mathbf{W}^H\cdot\mathrm{S}_{rc} &= [\mathbf{W}_1^H, \mathbf{W}_2^H]^T\cdot A_{P_0}\mathbf{V}\cdot\mathbf{A}\\ &= \begin{bmatrix}\mathbf{W}_1^H\cdot\mathbf{V}\\ \mathbf{W}_2^H\cdot\mathbf{V}\end{bmatrix}\cdot A_{P_0}\mathbf{A} = \mathbf{I}\cdot\mathbf{A} = A_{P_0}\mathbf{A}.\end{aligned} \tag{21}$$

By modulating (21), the scattering coefficients A_{P_1} and A_{P_2} of the symmetric point targets P_1 and P_2 can be obtained. The performance of the aforementioned beamforming-based left-right Doppler ambiguity resolving is mainly influenced by the difference between the steering vectors of the symmetrical Doppler components. With the increase in the number of array elements, the ambiguity suppression will be generally improved by the beamforming filter, especially on the forward-looking central area.

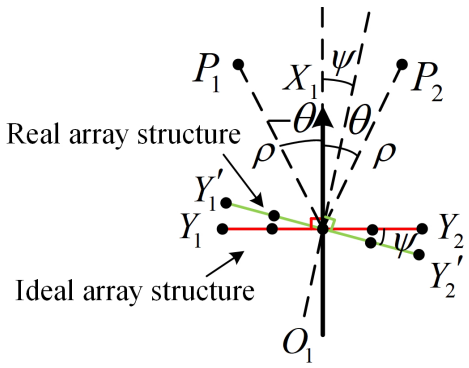


Fig. 4. Array structure with array deviation angle.

III. Adaptive Space–Time Array Deviation Angle Calibration

A. Array Structure With Array Deviation Angle

In practical applications, the array deviation angle, which can cause the mismatch between the azimuth angle and Doppler frequency, should be taken into account. The model of the array structure with an array deviation angle is established. As shown in Fig. 4, O_1X_1 denotes the nominal trajectory. And Y_1Y_2 is the ideal array structure, which is perpendicular to the trajectory O_1X_1. Due to the array deviation angle, the real array structure is turned to $Y_1'Y_2'$. The angle ψ between the real array structure and the ideal array structure is the array deviation angle.

P_1 and P_2 denote two arbitrary targets, which are symmetric about the X-axis. Their positions are $P_1(\rho, -\theta)$ and $P_2(\rho, \theta)$, respectively. Relative to the ideal array antenna, the steering vectors of the targets can be expressed as follows:

$$V_{P_1} = V(-\theta) = \exp\left(j\frac{2\pi}{\lambda}y_i\sin(-\theta)\right) \tag{22}$$

$$V_{P_2} = V(\theta) = \exp\left(j\frac{2\pi}{\lambda}y_i\sin(\theta)\right). \tag{23}$$

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Considering the array deviation angle, the steering vector of the targets related to the real array antenna are

$$\tilde{V}_{P_1} = V(-\theta - \psi) = \exp\left(j\frac{2\pi}{\lambda} y_i \sin(-\theta - \psi)\right) \quad (24)$$

$$\tilde{V}_{P_2} = V(\theta - \psi) = \exp\left(j\frac{2\pi}{\lambda} y_i \sin(\theta - \psi)\right). \quad (25)$$

Meanwhile, the real steering vector matrix $\tilde{\mathbf{V}}$ should be

$$\tilde{\mathbf{V}} = \begin{bmatrix} \tilde{V}_{P_1} \\ \tilde{V}_{P_2} \end{bmatrix} = \begin{bmatrix} V(-\theta - \psi) \\ V(\theta - \psi) \end{bmatrix}. \quad (26)$$

The two point targets P_1 and P_2, which are symmetric about the trajectory, still have the problem of Doppler ambiguity. However, the array deviation angle introduces errors in the steering vector, which results in a mismatch between azimuth angle and Doppler frequency in space–time characteristics.

B. Array Deviation Angle Calibration by Space–Time Characteristic Correction

Due to the existence of the array error, there is a mismatch between azimuth angle and Doppler frequency in space–time characteristic. The reason is the existence of the array deviation angle ψ, which causes an angle error between the real array distribution direction and the ideal array distribution direction, and further affects the beamforming-based left-right Doppler ambiguity resolving. Aiming at the matching problem of azimuth angle and Doppler frequency of the symmetric targets, this article proposes an array deviation angle calibration method using space–time characteristic correction.

Assuming that r is the reference slant range of the target and θ is the beam pointing azimuth angle of the target, the target imaging result of each channel can be expressed as

$$\mathrm{F}(r,\theta)_{K\times 1} = \left[\mathrm{S_{rc}}(\hat{t}, f_d; r, \theta, y_1)\ \mathrm{S_{rc}}(\hat{t}, f_d; r, \theta, y_2) \cdots \mathrm{S_{rc}}(\hat{t}, f_d; r, \theta, y_K)\right]^{\mathrm{T}} \quad (27)$$

where f_d is the Doppler frequency of the target expressed as

$$f_d = \frac{2\mathbf{v}\cdot\hat{\mathbf{r}}_{\mathbf{P}}}{\lambda|\hat{\mathbf{r}}_{\mathbf{P}}|} = \frac{2v\sin\alpha\cos\theta}{\lambda}. \quad (28)$$

From (28), we conclude that f_d a function of the beam pointing azimuth angle θ. In the $f_d - \theta$ plane, the space–time spectrum is shown in Fig. 5. The ideal space–time spectrum is drawn by the blue curve. Considering the array deviation angle, the red curve represents the real space–time spectrum. Assume a point target with the azimuth angle θ. According to (28), the Doppler frequency f_d can be obtained. This point target can be shown by point P in Fig. 5. A straight line F perpendicular to the f_d axis passes through the point P and intersects the real space–time spectrum at point $\tilde{P}$. θ, the azimuth angle coordinate of the point P, denotes the ideal azimuth angle. $\tilde{\theta}$, the azimuth angle coordinate of the point $\tilde{P}$, denotes the real azimuth angle. The array deviation angle, ψ, corresponds to the difference between θ and $\tilde{\theta}$. Thus, by measuring the difference between θ and $\tilde{\theta}$, the estimation of the array deviation angle can be performed.

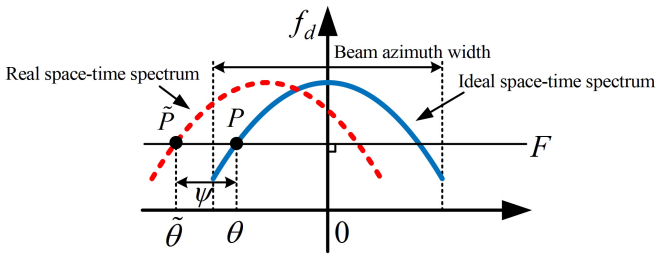


Fig. 5. Space–time spectrum with array deviation angle.

According to (28), the angle θ can be obtained from the Doppler frequency according to the ideal space–time characteristic

$$\theta = \arccos\frac{f_d\lambda}{2v\sin\alpha}. \quad (29)$$

In order to achieve the estimation of the angle $\tilde{\theta}$, the imaging area is divided into $\hat{N}$ units, and the steering vector corresponding to each unit is

$$\tilde{\mathbf{v}}(\theta_i)_{K\times 1} = \left[\exp\left(\tfrac{2\pi}{\lambda} y_1 \sin(\theta_i)\right)\ \exp\left(\tfrac{2\pi}{\lambda} y_2 \sin(\theta_i)\right) \cdots\ \exp\left(\tfrac{2\pi}{\lambda} y_K \sin(\theta_i)\right)\right]^T \quad \theta_i = \theta_1, \theta_2, \cdots, \theta_{\hat{N}}. \quad (30)$$

Considering that an array of K elements is excited by $\hat{N}$ target sources, the targets imaging result can be expressed more compactly as

$$\mathrm{F}(r,\theta)_{K\times 1} = \tilde{\mathbf{V}}(\Theta)_{K\times\hat{N}} \cdot \mathbf{S}_{\hat{N}\times 1} + \mathbf{n}_{K\times 1} \quad (31)$$

where $\mathbf{S}_{\hat{N}\times 1}$ is the scattering coefficient vector of the target for $\hat{N}$ units, and $\tilde{\mathbf{V}}(\Theta)_{K\times\hat{N}}$ is the steering vector matrix of the $\hat{N}$ units, which can be expressed as

$$\tilde{\mathbf{V}}(\Theta)_{K\times\hat{N}} = \left[\tilde{\mathbf{v}}(\theta_1)\ \tilde{\mathbf{v}}(\theta_2) \cdots \tilde{\mathbf{v}}\left(\theta_{\hat{N}}\right)\right]. \quad (32)$$

For FLMC-SAR imaging, the DBS algorithm has been performed to obtain the ambiguous imaging result. Thus, each imaging grid of the ambiguous imaging results contains two symmetric target sources. A maximum likelihood is applied to provide the estimation of the azimuth angle from the real space–time spectrum. Assuming that there are L samples $F_1, F_2, \cdots F_L$ separately submitted to the same Gaussian distribution with a mean of zero, the probability density function (PDF) is expressed as

$$f(x) = \frac{1}{\pi^K|\mathbf{R}|}\exp\left(-x^H\mathbf{R}^{-1}x\right) \quad (33)$$

where $\mathbf{R}$ is the covariance matrix of samples. Let $\mathbf{F} = [F_1, F_2, \cdots F_L]$, then we can derive the likelihood function of the unknown parameters Θ

$$f(\mathbf{F}|\Theta) = \frac{1}{\pi^{KL}|\mathbf{R}|^L}\exp\left(-\sum_{i=1}^{L} x^{iH}\mathbf{R}^{-1}x^i\right). \quad (34)$$

Then, the log-likelihood function can be obtained by applying the logarithm to (34)

$$L(\Theta) = -L\left[\ln|\mathbf{R}| + tr\left(\mathbf{R}^{-1}\mathbf{F}'\mathbf{F}\right)\right]. \quad (35)$$

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The maximum likelihood estimation of Θ is derived as [40]

$$\hat{\Theta} = \arg_{\Theta} \max L(\Theta) = \arg_{\Theta} \max\{tr[\mathbf{P} \cdot \mathbf{R}]\} \quad (36)$$

$$\mathbf{P} = \tilde{\mathbf{V}}(\Theta)\left(\tilde{\mathbf{V}}(\Theta)^H \cdot \tilde{\mathbf{V}}(\Theta)\right)^{-1}\tilde{\mathbf{V}}(\Theta)^H \quad (37)$$

where $tr[\cdot]$ is the trace of the matrix. $\mathbf{P}$ is the projection operator onto the space spanned by the columns of the matrix. Make a grid search to obtain the maximum peak of Θ and thus get the azimuth angle from the real space–time spectrum $\hat{\theta}$. With the estimation of angle $\hat{\theta}$, we can get the real space–time spectrum. According to the space–time spectrum shown in Fig. 4, the estimation of the array deviation angle $\hat{\psi}$ can be expressed as follows:

$$\hat{\psi} = \left||\hat{\theta}| - |\theta|\right|. \quad (38)$$

Considering the array deviation angle, the corrected steering vector of the ambiguous targets is obtained via the real space–time spectrum

$$\hat{V}_{P_1} = V\left(-\theta - \hat{\psi}\right) = \exp\left(j\frac{2\pi}{\lambda}y_i \sin\left(-\theta - \hat{\psi}\right)\right) \quad (39)$$

$$\hat{V}_{P_2} = V\left(\theta - \hat{\psi}\right) = \exp\left(j\frac{2\pi}{\lambda}y_i \sin\left(\theta - \hat{\psi}\right)\right). \quad (40)$$

Meanwhile, the compensated steering vector matrix $\hat{\mathbf{V}}$ should be

$$\hat{\mathbf{V}} = \begin{bmatrix} \hat{V}_{P_1} \\ \hat{V}_{P_2} \end{bmatrix} = \begin{bmatrix} V\left(-\theta - \hat{\psi}\right) \\ V\left(\theta - \hat{\psi}\right) \end{bmatrix}. \quad (41)$$

With the compensated steering vector matrix $\hat{\mathbf{V}}$ adopted in beamforming, the algorithm proposed in this article can be used to solve left-right Doppler ambiguity in FLMC-SAR imaging.

For the clarity of the proposed method for imaging and ambiguity removal for FLMC-SAR, we have employed a feasible procedure as follows.

The first step is utilizing the DBS algorithm in the time domain to obtain the ambiguous image of each channel. The high-resolution imaging provided by the time domain synthetic aperture resources is a prerequisite for the left-right Doppler ambiguity removal and the array deviation angle calibration.

The second step is the array deviation angle calibration. In this step, we first select samples for each range bin, which is similar to the procedure in the conventional phase gradient autofocus (PGA). Then, the space–time characteristic correction is performed via maximum likelihood estimation. With the estimation of the array deviation angle, the array is calibrated by correcting the mismatch between the azimuth angle and Doppler frequency. Moreover, since the array deviation angle presents a range-varying characteristic, the proposed algorithm adopts the data-driven adaptive processing method in every range unit to realize the array deviation angle calibration.

The third step is removing the left-right Doppler ambiguity via beamforming. The LCMV algorithm is adopted in beamforming-based left-right Doppler ambiguity resolving under the constraint of maintaining the component of the filtered signal in the desired target direction and forming a null in the ambiguous direction. However, if the array deviation angle exists, the ambiguous direction will move outside the null, resulting in serious performance degradation. Thus, the array deviation angle calibration is especially important for the left-right Doppler ambiguity resolving.

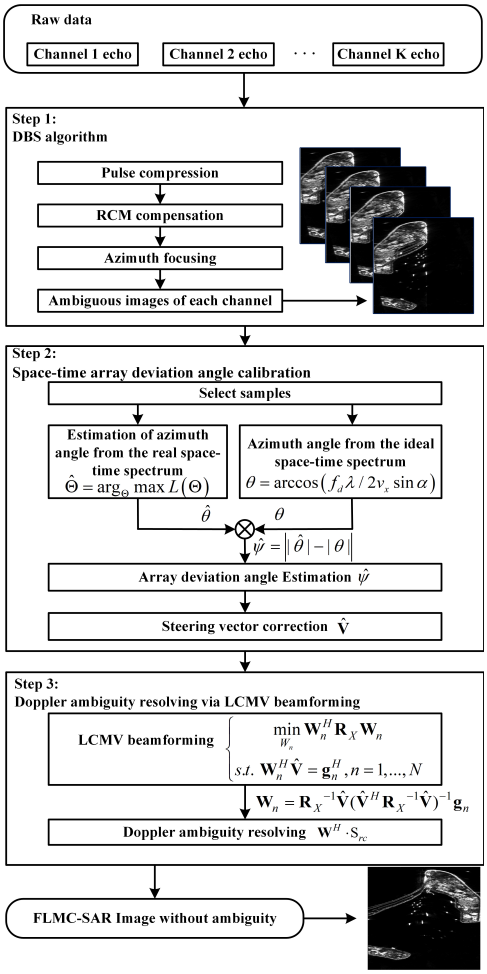


Fig. 6. Overall FLCM-SAR imaging flowchart integrated with array deviation angle calibration.

In the following section, with the array deviation angle calibration, the FLMC-SAR can achieve forward-looking unambiguity imaging, which indicates the efficiency of the proposed algorithm. And, the overall FLCM-SAR imaging flowchart integrated with the array deviation angle calibration is shown in Fig. 6.

C. Computation Cost

The proposed approach runs in a three-step process.

Step 1—DBS Algorithm: To perform SAR imaging with $\tilde{N}$ range bins and M azimuth bins of K channels, fast Fourier transform FFT) is carried out M times with respect to $\tilde{N}$ range bins to achieve the range compression, and the RCM compensation can be obtained by carrying out FFT $\tilde{N}$ times with respect to M azimuth bins. Then the azimuth focusing can be achieved by performing FFT $\tilde{N}$ times with respect to M azimuth bins. So the total computation of DBS algorithm would be $O\left(K\left(M\log\tilde{N} + 2\tilde{N}\log M\right)\right)$.

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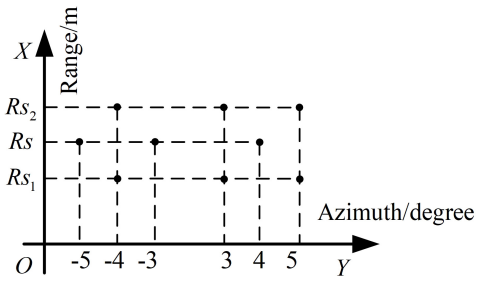


Fig. 7. Geometric distribution of the targets.

Step 2—Array Deviation Angle Calibration: To achieve the array deviation angle calibration, we need carry out maximum likelihood estimation $\tilde{N}$ times. Assuming that the imaging area is divided into $\hat{N}$ units, the maximum likelihood estimation requires $\tilde{N}$ inversions for the matrix of $\hat{N}/2 \times \hat{N}/2$. The total computation of the array deviation angle calibration would be $O\left(\tilde{N}\left((\hat{N}/2)\right)^3\right)$.

Step 3—Beamforming-Based Left-Right Doppler Ambiguity Resolving: In the beamforming processing with $\tilde{N}$ range bins and M azimuth bins of K channels, we operate $M/2$ inversions for the matrix of $K \times K$. The total computation of resolving the beamforming-based left-right Doppler ambiguity would be $O\left((M/2)K^3\right)$.

Therefore, the total computation of the proposed method is

$$\begin{aligned} &O\left(K\left(M\log\tilde{N} + 2\tilde{N}\log M\right)\right) \\ &\quad + O\left(\tilde{N}\left((\hat{N}/2)\right)^3\right) + O\left((M/2)K^3\right) \\ &\quad = O\left(KM\log\tilde{N} + K\tilde{N}\log M + \tilde{N}\hat{N}^3 + MK^3\right). \end{aligned}$$

IV. Experiments and Discussion

In order to demonstrate the performance and effectiveness of the proposed approach for forward-looking SAR imaging, we perform different experiments to show its superiority from different aspects in this section. The experiment of the FLMC-SAR system with array deviation angle is carried out. The experiment contains three cases, as follows.

A. Case 1: Point Target Simulation

To verify the effectiveness of the proposed algorithm for FLMC-SAR imaging, we compare the imaging results of the point targets after the array deviation angle calibration with those without the array deviation angle calibration. As shown in Figs. 7 and 9, point targets are placed in the imaging area. Here, the X-axis denotes the target reference slant range and the Y-axis denotes the target azimuth angle. We set the SNR to be 10 dB, and the relevant simulation parameters are shown in Table I. The array error, which is expressed as an array deviation angle between the real array structure and the ideal one, is added into the echo.

Fig. 8 shows the point targets' imaging results. Fig. 8(a)–(d) indicates the ambiguous imaging results, ambiguity resolving results without the array deviation angle, ambiguity resolving results with an array deviation angle, and ambiguity resolving results with an array deviation angle via the adaptive space–time array deviation angle calibration, respectively. One can see that the point targets can be distinguished clearly. But the imaging results of scenes on both sides of the trajectory are aliased, caused by the left-right Doppler ambiguity as shown in Fig. 8(a). Without the array deviation angle, beamforming can solve the left-right Doppler ambiguity effectively, as shown in Fig. 8(b). However, the images suffer from the array deviation angle. In some areas highlighted in Fig. 8(c), the array deviation angle leads to the failure of eliminating the left-right Doppler ambiguity. In order to carry out the array deviation angle calibration, the samples with large scattering coefficients are selected in the ambiguous imaging results. Highlighted in Fig. 8(a), there are three targets selected for the space–time characteristic correction. And the array deviation angle calibration results are shown in Table II. As can be seen, the different range unit performs the different compensation of the array deviation angle, which indicates that the array deviation angle calibration needs to consider the space-varying characteristic of the array deviation angle.

The LCMV beamforming response pattern is shown in Fig. 9. The black curve is the LCMV beamforming response pattern without an array deviation angle calibration. And the black circle is the array gain in the ambiguous direction without array deviation angle calibration. It is notable that the array gain in the ambiguous direction is high which is not conducive to ambiguity suppression. With the compensation of the array deviation angle, the LCMV beamforming response pattern is drawn by the red curve in Fig. 9. The red circle marks the array gain in the ambiguous direction after array deviation angle calibration, and the gain value rapidly decreases to zero. Through the array deviation angle calibration, the steering vector is corrected. And the LCMV beamforming forms a null in the ambiguous direction, which causes the array gain in the ambiguous direction to drop dramatically. Therefore, left-right ambiguity resolving is successful. The ambiguity resolving result via adaptive space–time array deviation angle calibration is shown in Fig. 8(d). The azimuth ambiguity-to-signal ratio (AASR) [41] is applied as an evaluation factor of the azimuth

TABLE I
Simulation Parameters

Parameter	Value	Parameter	Value
Carrier frequency	25GHz	PRF	8000Hz
Bandwidth	55MHz	Beam azimuth width	22degree
Number of array elements	5	Beam pitch width	22degree
Platform height	4000m	Real aperture length	0.16m

TABLE II
Array Deviation Angle Calibration Result

	Ideal azimuth angle θ	Estimation of real azimuth angle $\hat{\theta}$	Estimation of array deviation angle $\hat{\psi}$
Target 1	5 Degree	-6.07 Degree	1.07 Degree
Target 2	4 Degree	-5.19 Degree	1.19 Degree
Target 3	3 Degree	-4.31 Degree	1.37 Degree

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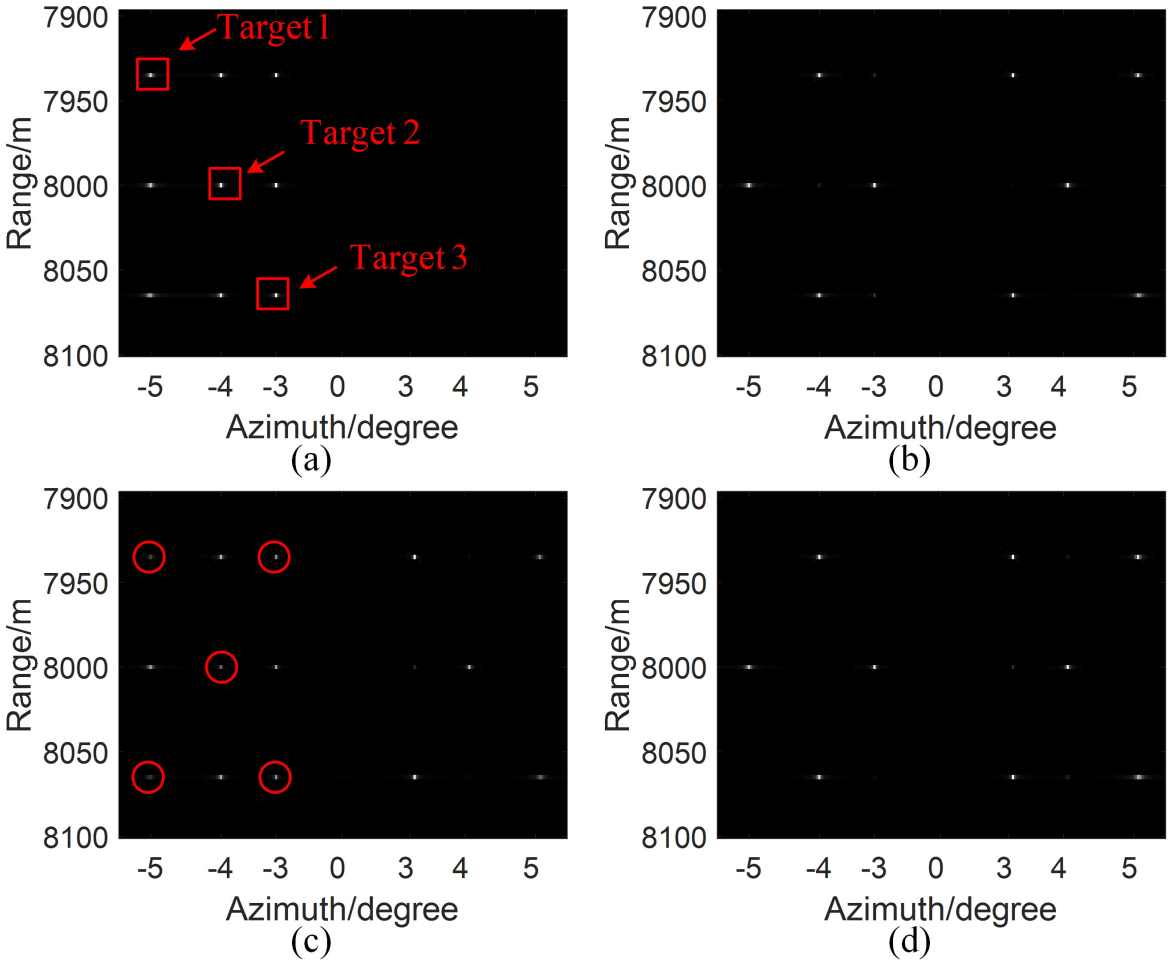


Fig. 8. Point targets' Imaging results. (a) Ambiguous imaging results. (b) Ambiguity resolving results without an array deviation angle. (c) Ambiguity resolving results with an array deviation angle. (d) Ambiguity resolving result with an array deviation angle via adaptive space–time array deviation angle calibration.

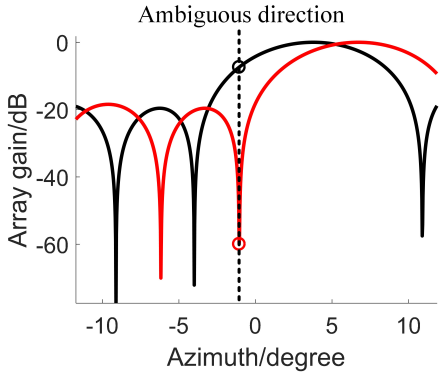


Fig. 9. LCMV beamforming response pattern. Black curve: LCMV beamforming response pattern without an array deviation angle calibration. Red curve: LCMV beamforming response pattern with array deviation angle calibration. Black circle: array gain in the ambiguous direction without array deviation angle calibration. Red circle: array gain in the ambiguous direction with array deviation angle calibration.

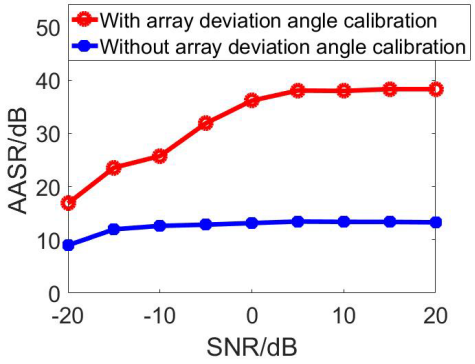


Fig. 10. AASR of the point targets imaging result.

ambiguity of the FLMC-SAR image. And the AASR of Fig. 8(d) as a function of the SNR is also shown in Fig. 10 in order to show the robustness of the calibration algorithm. With array deviation angle calibration, significant improvement in left-right Doppler ambiguity resolving at moderate SNR is ensured.

B. Case 2: Surface Target Simulation

In this section, the simulation experiment of a surface target is carried out. We set the SNR to be 10 dB, and the relevant simulation parameters are shown in Table I. The simulation uses the SAR image obtained by real data as a reference image, as shown in Fig. 11(a). The left-right Doppler ambiguity still appears in this experiment, and the ambiguous imaging result is shown in Fig. 11(b). In order to show the influence of the array deviation angle on resolving the beamforming-based

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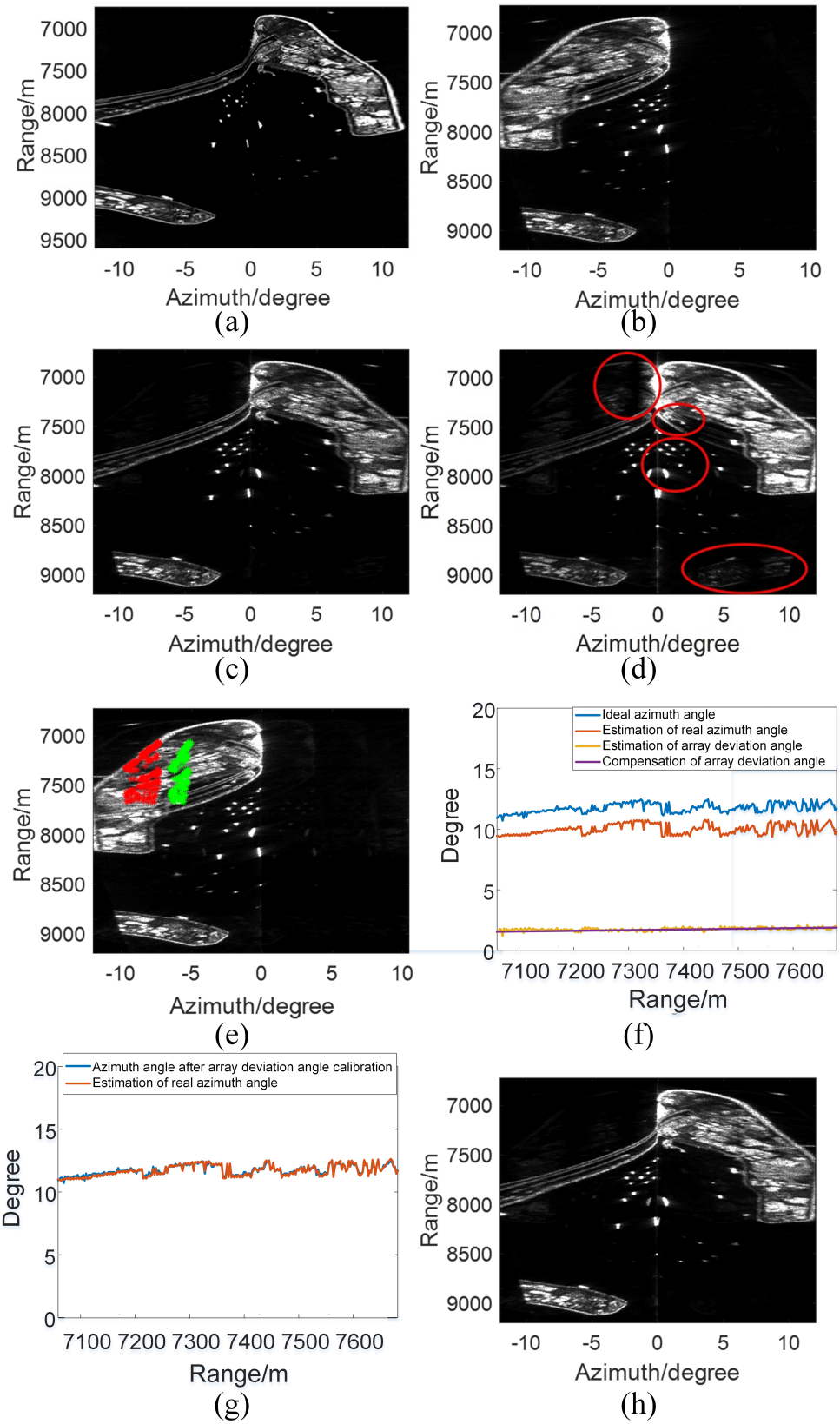


Fig. 11. FLMC-SAR images of the surface target. (a) Reference image. (b) Ambiguous imaging result. (c) Imaging result without an array deviation angle. (d) Imaging result with an array deviation angle. (e) Selected samples and space–time spectrum characteristic of the samples. (f) Comparison between the ideal azimuth angle, estimation of the real azimuth angle, and estimation of array deviation angle. (g) Comparison between azimuth angle after array deviation angle calibration and estimation of the real azimuth angle. (h) Imaging result with array deviation angle calibration by the proposed algorithm.

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left-right Doppler ambiguity, we give a comparison of imaging results shown in Fig. 11(c) without an array deviation angle and that shown in Fig. 11(d) with an array deviation angle degree. As stated earlier, the array deviation angle leads to serious performance degradation of the beamforming-based left-right Doppler ambiguity resolving. In some areas highlighted in Fig. 11(d), it leads to the inability to solve the left-right Doppler ambiguity.

For the left-right Doppler ambiguity resolving in the practical applications, the array deviation angle calibration is carried out via the proposed method. First, we select samples in the ambiguous imaging result to perform space–time characteristic correction. Moreover, in the point target simulation experiment, we can perform array deviation angle calibration one by one for each range unit containing the target, because point target is isolated and does not have the problem of discontinuity. However, in the surface target simulation, to ensure the continuity of the target, we need to fit the array deviation angle with a smooth curve. Since the varying characteristics of the array deviation angle are caused by the different radar's light of sight of different range units, we can linearly treat the range-varying characteristic. The dominant scatters highlighted by the red points in Fig. 11(e) are selected, which are several range units of some buildings with a large scattering coefficient. And the space–time characteristic estimation of the samples is shown in Fig. 11(e) highlighted by the green points. The comparison between the ideal azimuth angle θ, estimation of the real azimuth angle $\hat{\theta}$, and estimation of array deviation angle $\hat{\psi}$ is shown in Fig. 11(f). It is clear that there is a range variance in the array deviation angle. For the array deviation angle calibration, we need to fit this angle error and perform compensation. After the compensation, the comparison between the azimuth angle after the array deviation angle calibration and the estimated azimuth angle from the real space–time spectrum is shown in Fig. 11(g). And, the imaging result of the beamforming with the array deviation angle calibration obtained by the proposed algorithm is shown in Fig. 11(h). The AASR of the surface targets as a function of SNR is shown in Fig. 12. With the moderate SNR, the highlighted area with the left-right Doppler ambiguity in Fig. 11(d) can achieve imaging without ambiguity by the proposed method. Under the Inter (R)Core (TM) i7-6700HQ CPU MatlabR2016b, we calculate the operation time. The DBS algorithm takes the longest time, 7.77 s, followed by the array deviation angle calibration, 4.75 s, with the beamforming taking the least, 4.28 s. The results align with the trend of the theoretical computation cost.

C. Case 3: Real Data Experiment

In the following section, the real data experiment is carried out to investigate the performance of the proposed method on FLMC-SAR imaging. The real data are collected by a K-band experimental FLMC-SAR system. The radar is mounted on-board an airborne platform, working in spotlight mode with 8° beam azimuth width and 20° beam pitch width. The pulse repetition frequency (PRF) was 6000 Hz, the synthetic aperture times was about 1.2 s containing 8192 pulses.

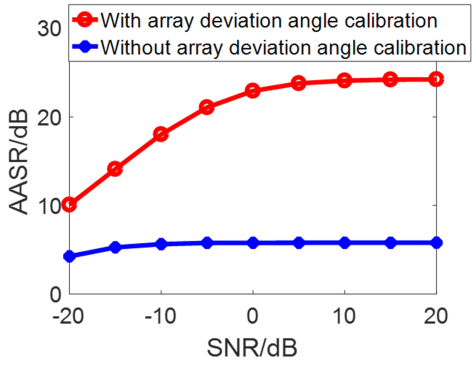


Fig. 12. AASR of the surface targets imaging result.

TABLE III
SIMULATION PARAMETERS

Parameter	Value	Parameter	Value
Carrier frequency	20GHz	PRF	6000Hz
Bandwidth	50MHz	Beam azimuth width	20degree
Number of array elements	5	Beam pitch width	22degree
Platform height	4000m	Real aperture length	0.16m

The height of the platform is 4000 m, and the reference slant range of the center of the scene is 8000 m. The system is equipped with a five-channel array antenna with a real aperture of 0.16 m. A high-precision position and orientation system provides trajectory information during the data collection. The relevant experiment parameters of the system are shown in Table III.

The imaging results of real data are shown in Fig. 13. The horizontal direction is the azimuth bin, and the vertical direction is the range bin. The ambiguous imaging result is shown in Fig. 13(a). Due to the effect of platform vibration, the compensation of the motion errors and autofocus [33], [34] is of great significance to achieve a well-focused image. After the motion compensation and autofocusing, the imaging results are shown in Fig. 13(b). Apparently, the images on both sides of the trajectory generated by the DBS algorithm are mixed due to the left-right Doppler ambiguity. The imaging result after beamforming applied to the FLMC-SAR Doppler resolving without the array deviation angle calibration is shown in Fig. 13(c). It can be noted that some areas that are highlighted in Fig. 13(c) are seriously blurred. In order to improve the performance of the beamforming-based left-right Doppler ambiguity resolving, an adaptive space–time array deviation angle calibration is performed for FLMC-SAR Imaging. The dominant scatters highlighted in Fig. 13(d) by the red points are selected as the samples for estimating the array deviation angle. And the space–time characteristic estimation of the samples is shown in Fig. 13(d) by the green points. The comparison between the ideal azimuth angle θ, estimation of the real azimuth angle $\hat{\theta}$, and estimation of array deviation angle $\hat{\psi}$ is shown in Fig. 15(e). We can see that there is a range-varying error between θ and $\hat{\theta}$. For calibrating the array deviation angle, we also need to linearly fit this angle error and perform compensation. After the compensation, the comparison between the azimuth angle

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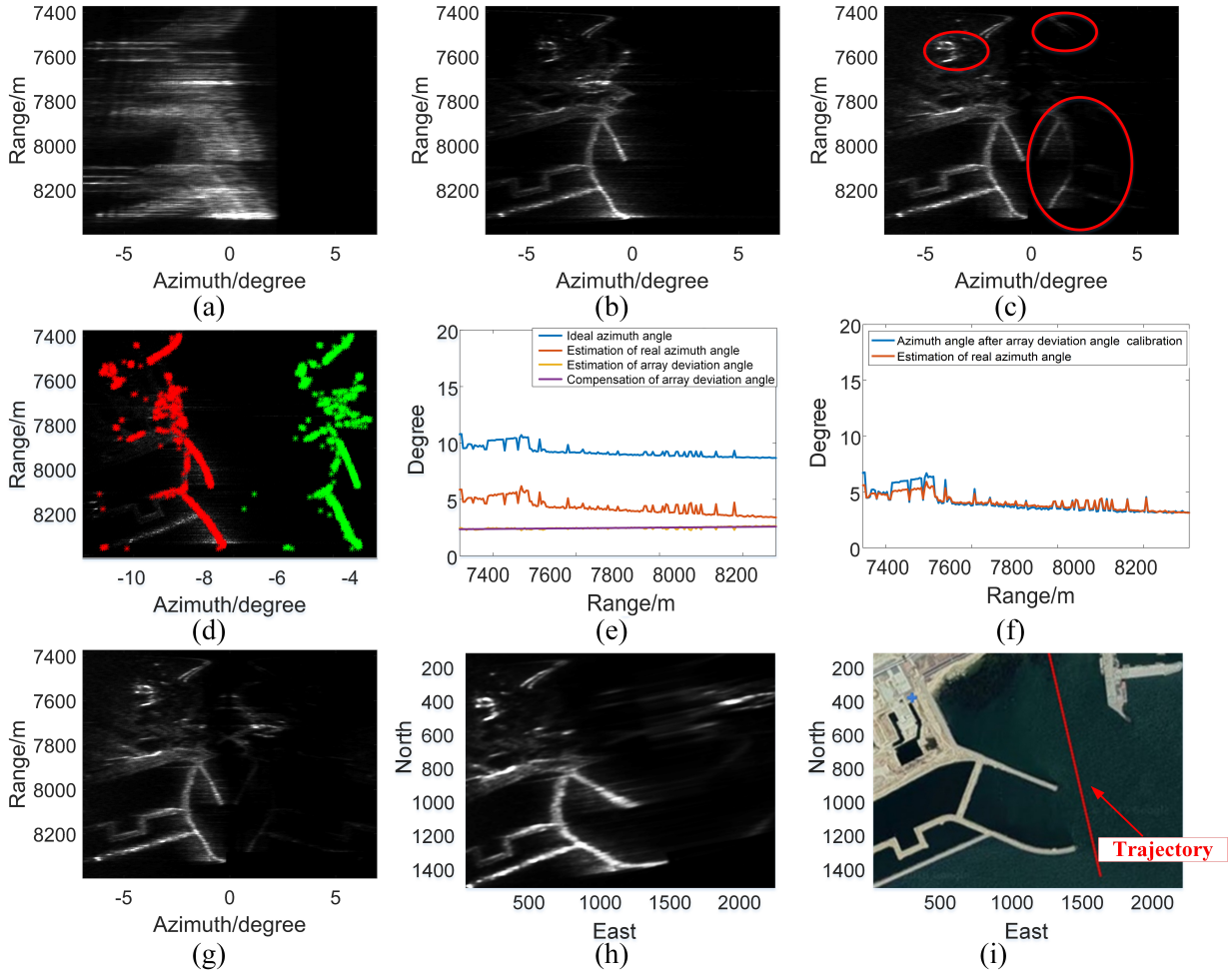


Fig. 13. FLMC-SAR images of real data. (a) Without motion compensation and autofocus. (b) With motion compensation and autofocus. (c) Beamforming result without array deviation angle calibration. (d) Selected samples and space–time characteristic estimation of the samples. (e) Comparison between the ideal azimuth angle, estimation of the real azimuth angle, and estimation of array deviation angle. (f) Comparison between azimuth angle after array deviation angle calibration and estimation of the real azimuth angle. (g) Imaging result of real data obtained by the proposed algorithm. (h) Projection of imaging results in the geographic coordinate system. (i) Satellite image of the imaging area.

after the array deviation angle calibration and the estimation of the azimuth angle from the real space–time spectrum is shown in Fig. 13(f). And, the imaging result of the real data with the array deviation angle calibration obtained by the proposed algorithm is shown in Fig. 13(g). The projection of Fig. 13(g) in the geographic coordinate system is shown in Fig. 13(h). The satellite image is shown in Fig. 13(i), and the red line is the trajectory. Compared to the satellite image, the image generated by the proposed adaptive space–time array deviation angle calibration and the imaging algorithm with ambiguity resolving for FLMC-SAR is well-focused. This real data experiment validates the effectiveness of the proposed algorithm. Under the Inter (R)Core (TM) i7-6700HQ CPU MatlabR2016b, we calculate the operation time. The DBS algorithm takes the longest time, 9.86 s, followed by the array deviation angle calibration, 6.43 s, with the beamforming taking the least, 5.36 s. The results align with the trend of the theoretical computation cost.

V. Conclusion

In this article, an adaptive space–time array deviation angle calibration and imaging algorithm has been developed for the FLMC-SAR system. We can perform the DBS algorithm in the time domain. Then, the array deviation angle calibration is performed in the spatial domain to realize the space—time characteristic correction. It has been found that beamforming as an important tool to solve the left-right Doppler ambiguity problem in forward-looking imaging reacts very sensitively to the mismatch of azimuth angle and Doppler frequency. With the accurate matching between azimuth angle and Doppler frequency, beamforming-based left-right Doppler ambiguity resolving has shown good performance. Simulation and real data experiments validate the effectiveness of the proposed algorithm for the FLMC-SAR system equipped with a high-precision navigation system. It needs to be emphasized that this article only deals with the space–time mismatch resulting from the array deviation angle. Furthermore, residue motion error will induce Doppler shift, leading to degradation of Doppler ambiguity removal, especially when the navigation system is not accurate. The term “array attitude calibration” is our future work and imagery for the FLMC-SAR system with a low-precision navigation system is a significant expansion. What is more, the single-transmitter multiple-receiver array is used in the FLMC-SAR system. Of course, the multiple-input

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multiple-output (MIMO) array is not only feasible but also will introduce some potential improvements. MIMO configuration for FLMC-SAR will be investigated in our further work.

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